Microprocessors and Embedded Systems - Lecture Notes

1. Microprocessor Architecture

A microprocessor executes instructions in a fetch-decode-execute cycle. The program counter holds the address of the next instruction, the instruction register holds the instruction being decoded, and the accumulator holds intermediate results. The arithmetic logic unit (ALU) performs arithmetic and logic operations, and the status (flag) register records conditions such as zero, carry and overflow.

2. The 8051 Microcontroller

The 8051 is an 8-bit microcontroller with 4 KB of on-chip program ROM and 128 bytes of RAM. It has 32 general-purpose I/O pins arranged in four 8-bit ports (P0 to P3), two 16-bit timers (Timer 0 and Timer 1), and one serial port (UART). It uses 40 pins in a dual in-line package (DIP40).

Timer modes: Mode 0 is 13-bit, Mode 1 is 16-bit, Mode 2 is 8-bit auto-reload and Mode 3 is split timer. The machine cycle of the 8051 is 12 oscillator periods; with a 12 MHz crystal the machine cycle is 1 microsecond.

3. Memory and Addressing

Von Neumann architecture uses a single memory space for code and data; Harvard architecture uses separate program and data memories. The 8051 uses a Harvard architecture. Addressing modes include immediate, direct, register, register indirect and indexed (used with the data pointer DPTR for look-up tables).

Cache memory sits between the CPU and main memory to reduce average access time. Common cache mapping schemes are direct-mapped, fully associative and set associative.